

ALGORITHMS & COMPLEXITY

LECTURE 7

We present a method called

the method of generating functions (GF)
which may apply to:

- non-linear difference scheme
- schemes of non-fixed order
- convolved schemes having more than one recursive scheme

It has one main weakness - to be explained later.

For $\{a_n\}_{n \geq 0} \xrightarrow{\text{mapping}} f(x) = \sum_{n=0}^{\infty} a_n x^n$ (□)
function series

$$f(x) = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_k x^k$$

We know from the calculus that
for such power series

$\exists R \geq 0 \quad \forall x \in (-R, R) \quad \sum_{n=0}^{\infty} a_n \overbrace{x^n}^{\text{number series}}$ is convergent

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R - called radius of convergence
can be $0 < R < \infty$ or $R = \infty$

(then only for $\bar{x} = 0$ we have convergence)
or $R = -\infty$ (then $\forall x \in \mathbb{R}$ we have convergence)

In general if $\lim_n \left| \frac{a_{n+1}}{a_n} \right| = g_1$ (1)

or if $\lim_n \sqrt[n]{|a_n|} = g_2$ (2)

Then $R = \frac{1}{g}$ (if both g_1 & g_2 exist they satisfy $g_1 = g_2$)

Properties of $f(x) = \sum_{n=0}^{\infty} a_n x^n$ $x \in (-R, R)$
 $f'(x) = \sum_{n=1}^{\infty} a_n \cdot n x^{n-1}$

If $g_1(x) = \sum_{n=1}^{\infty} a_n x^n$, $x \in (-R_1, R_1)$

$g_2(x) = \sum_{n=1}^{\infty} \hat{a}_n x^n$, $x \in (-R_2, R_2)$

then

$$g_1(x) + g_2(x) = \sum_{n=1}^{\infty} (a_n + \hat{a}_n) x^n$$

$$x \in (-\min(R_1, R_2), \min(R_1, R_2))$$

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We illustrate method of generating functions in several examples.

Example:
$$(*) \begin{cases} a_n - 3a_{n-1} + 2a_{n-2} = 0 \\ a_0 = 0, \quad a_1 = 1 \end{cases}$$

Linear scheme of order 2, homogeneous with constant coefficients & with initial values $\Rightarrow$ it has a unique sol.

$\{a_n\}_{n \geq 0}$ satisfying (*)

Take $\{a_n\}_{n \geq 0} \longrightarrow f(x) = \sum_{k=0}^{\infty} a_k x^k$

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$$

$$\begin{aligned} -3xf(x) &= -3x \sum_{k=0}^{\infty} a_k x^k \\ &= -3a_0 x - 3a_1 x^2 - \dots - 3a_{n-1} x^n + \dots \end{aligned}$$

$$\begin{aligned} 2x^2 f(x) &= 2x^2 \sum_{k=0}^{\infty} a_k x^k \\ &= 2x^2 a_0 + 2a_1 x^3 + \dots + 2a_{n-2} x^2 + \dots \end{aligned}$$

Thus we have:

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$$f(x) - 3xf(x) + 2x^2 f(x)$$

$$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$- 3a_0 x - 3a_1 x^2 - 3a_2 x^3 - 3a_3 x^4 - \dots$$

$$+ 2a_0 x^2 + 2a_1 x^3 + 2a_2 x^4 + \dots$$

$$a_0 + a_1 x - 3a_0 x + x^2(a_2 - 3a_1 + 2a_0) + x^3(a_3 - 3a_2 + 2a_1)$$

$$+ x^4(a_4 - 3a_3 + 2a_2) + \dots + x^n(a_n - 3a_{n-1} + 2a_{n-2})$$

$$+ \dots$$

But all expressions in (brackets) vanish as a_n satisfies (*). Thus

$$(c) \boxed{f(x) - 3xf(x) + 2x^2 f(x) = a_0 + a_1 x - 3a_0 x}$$

Since $a_0 = 0$ & $a_1 = 1$
in the function equation (c) we
compute $f(x)$ to get:

$$\boxed{f(x) = \frac{x}{1 - 3x + 2x^2}}$$

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The general trick is now:

$$\underbrace{\sum_{n=1}^{\infty} a_n x^n}_{\text{as we assumed that } f(x) \text{ is built from } \{a_n\}_{n \geq 0} \text{ (see \blacksquare)}} = f(x) = \underbrace{\sum_{n=1}^{\infty} b_n x^n}_{\text{we know that each } f \text{ can be expanded by Taylor formula}}$$

as we assumed that $f(x)$ is built from $\{a_n\}_{n \geq 0}$ (see ■)

we know that each f can be expanded by Taylor formula

$$b_n = \frac{f^{(n)}(0)}{n!}$$

As f has unique series representation then

$$\boxed{a_n = b_n}$$

however there is a weakness in the above

- a_n is given recursively in (*)
- to compute $b_n = \frac{f^{(n)}(0)}{n!}$ we need recursively

$$f^{(n)}(0) = \left(f^{(n-1)}(0) \right)' \quad (***)$$

which is also relying on previous derivatives. And it is hard to see some general pattern

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- we do not want to replace one recursion (*) with other one (***) once we transform implicit form into explicit form.

Sometimes $\frac{f^n(0)}{n!}$ can be done without recursion, or with the recursion in which we see a general pattern which can be proved by mathematical induction. Indeed in our example this is the case:

$$f(x) = \frac{x}{1-3x+2x^2} \quad (\text{rational function})$$

We calculate Δ for denominator as we want to factorize it.

$$\Delta = 1 \quad x_1 = \frac{1}{2} \quad \& \quad x_2 = 1$$

$$1-3x+2x^2 = 2(x-\frac{1}{2})(x-1) = (2x-1)(x-1) \\ = (1-2x)(1-x)$$

$$\text{Thus } f(x) = \frac{x}{(1-x)(1-2x)}$$

Next step is to use the so-called
PARTIAL FRACTION DECOMPOSITION

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$$\begin{aligned}\frac{x}{(1-x)(1-2x)} &= \frac{A}{1-x} + \frac{B}{1-2x} \\ &= \frac{(1-2x)A + (1-x)B}{(1-x)(1-2x)} \\ &= \frac{x(-2A-B) + (A+B)}{(1-x)(1-2x)} \quad \forall x\end{aligned}$$

$$1 \cdot x + 0 = x(-2A-B) + (A+B) \quad \forall x$$

$\Downarrow$ two polynomials are the same if their coefficients are equal

$$\begin{cases} -2A - B = 1 \\ A + B = 0 \end{cases} \Rightarrow A = -B \Rightarrow \begin{cases} 2B - B = 1 \\ B = 1 \\ A = -1 \end{cases}$$

Hence $f(x) = \frac{-1}{1-x} + \frac{1}{1-2x}$

Taking into account that sum of geometric progression for $|q| < 1$ amounts to

$$\frac{1}{1-q} = 1 + q^1 + q^2 + q^3 + \dots + q^n + \dots$$

$$-\frac{1}{1-x} = -(1 + x + x^2 + x^3 + \dots + x^n + \dots)$$

$$\frac{1}{1-2x} = 1 + 2^1 x^1 + 2^2 x^2 + 2^3 x^3 + \dots + 2^n x^n + \dots$$

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Consequently

$$\begin{aligned} f(x) &= - \left(1 + x + x^2 + \dots + x^n + \dots \right) \\ &\quad + \left(1 + 2x + 2^2x^2 + \dots + 2^n x^n + \dots \right) \\ &= x + (2^2 - 1)x^2 + (2^3 - 1)x^3 + \dots + (2^n - 1)x^n + \dots \\ &= \underbrace{(2^0 - 1)}_0 x^0 + (2^1 - 1)x^1 + (2^2 - 1)x^2 + (2^3 - 1)x^3 \\ &\quad + \dots + (2^n - 1)x^n + \dots \\ &= \sum_{n=0}^{\infty} (2^n - 1)x^n \end{aligned}$$

So on one hand & on other hand

$$\sum_{n=0}^{\infty} a_n x^n = f(x) = \sum_{n=0}^{\infty} (2^n - 1)x^n$$

$$\left\{ \begin{array}{l} a_n = 2^n - 1 \\ a_0 = 0 \end{array} \right.$$

$$\boxed{a_n = \Theta(2^n)}$$

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We give the example in which the method of generating functions can be applied to the convolved system of 2 recursive schemes:

Example: Let in reactor of particles a_n — denotes the number of neutrons of high energy at time $t_n = n$ milliseconds passed from $t_0 = 0$

b_n — denotes the number of neutrons of low energy at time $t_n = n$ milliseconds passed from $t_0 = 0$

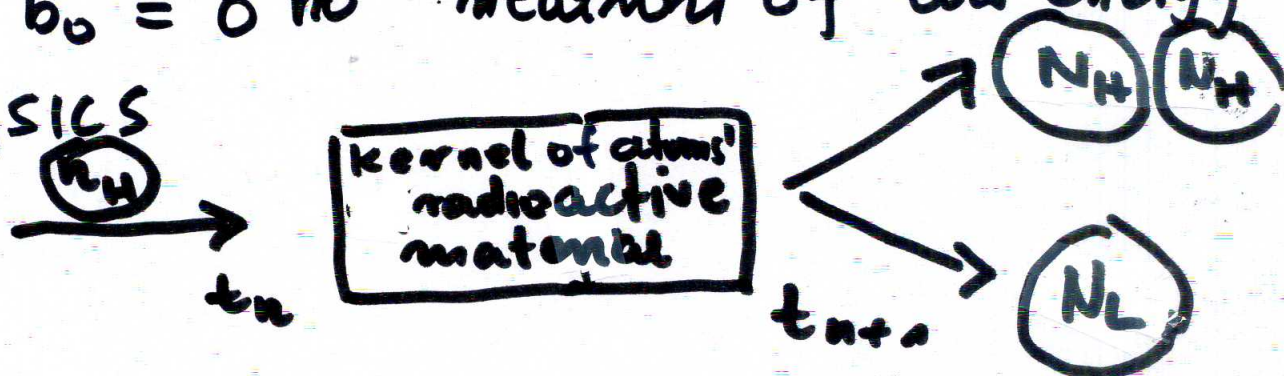
Assume that at $t_0 = 0$ in the reactor we "inject"

$a_0 = 1$ one neutron of high energy

$b_0 = 0$ no neutron of low energy

In physics

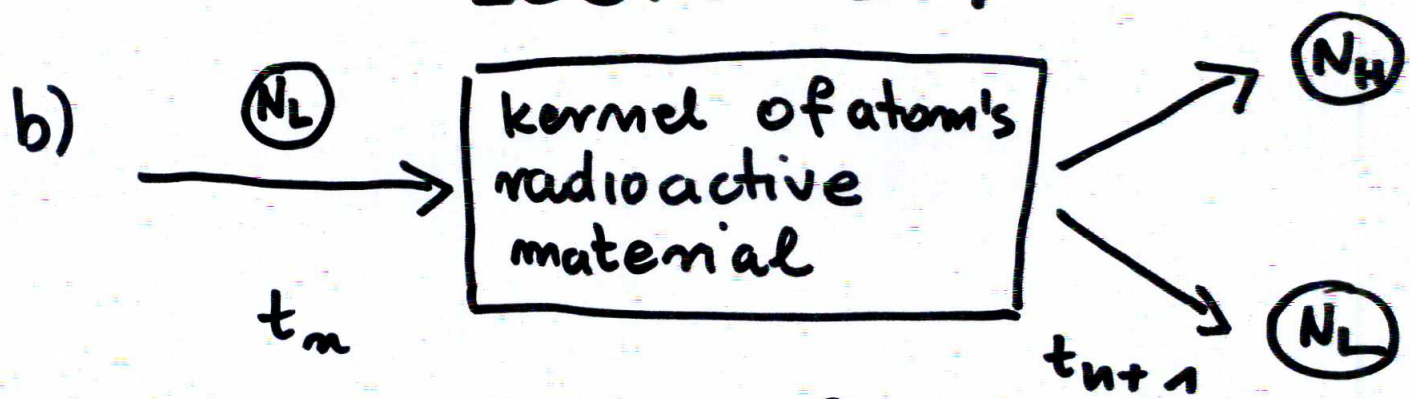
a)



high-energy neutron after hitting radioactive material produces 2 N_H & 1 N_L .
atoms' kernel — 9 —

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Low energy neutron (N_L) after hitting atoms' kernel of radioactive material produces 1 (N_H) & 1 (N_L) .

a) & b) have the following recurrent representation at time t_{n+1} :

$$a_{n+1} = \underset{\substack{\uparrow \\ \text{contribution of } a_n \text{ neutrons} \\ \text{at time } t_n}}{2a_n} + b_n \leftarrow \begin{array}{l} \text{contribution of} \\ b_n \text{ neutrons at } t_n \end{array}$$

$$b_{n+1} = \underset{\substack{\downarrow \\ \text{contribution of } a_n \text{ neutrons} \\ \text{at time } t_n}}{a_n} + b_n$$

So

$$(*) \quad \begin{cases} a_{n+1} = 2a_n + b_n, & a_0 = 1 \\ b_{n+1} = a_n + b_n, & b_0 = 1 \end{cases}$$

If we use $c_n = (a_n, b_n)$ scheme

* cannot be transformed into

$$c_{n+1} = \bar{K} \cdot c_n + \bar{K}_2$$

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$$\{a_n\}_{n \geq 0} \longrightarrow f(x) = \sum_{n=0}^{\infty} a_n x^n \quad (**)$$

$$\{b_n\}_{n \geq 0} \longrightarrow g(x) = \sum_{n=0}^{\infty} b_n x^n \quad (***)$$

$$a_{n+1} = 2a_n + b_n \quad | \cdot x^{n+1}$$

$$b_{n+1} = a_n + b_n \quad | \cdot x^{n+1}$$

$$\Downarrow$$

$$\begin{cases} a_{n+1} x^{n+1} = 2a_n x^{n+1} + b_n x^{n+1} \\ b_{n+1} x^{n+1} = a_n x^{n+1} + b_n x^{n+1} \end{cases}$$

$\Downarrow$

$$\begin{cases} \sum_{n=0}^{\infty} a_{n+1} x^{n+1} = 2 \sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} b_n x^{n+1} \\ \sum_{n=0}^{\infty} b_{n+1} x^{n+1} = \sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} b_n x^{n+1} \end{cases}$$

we take infinite sum

$\Downarrow$

$$n+1 = k$$

$$n=0 \Rightarrow k=1$$

$$n=\infty \Rightarrow k=\infty$$

$$\begin{cases} \sum_{k=1}^{\infty} a_k x^k = 2x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} b_n x^n \\ \sum_{k=1}^{\infty} b_k x^k = x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} b_n x^n \end{cases}$$

$$k=n \begin{cases} \sum_{n=0}^{\infty} a_n x^n - a_0 x^0 \stackrel{**}{=} 2x f(x) + x g(x) \\ \sum_{n=0}^{\infty} b_n x^n - b_0 x^0 \stackrel{***}{=} x f(x) + x g(x) \end{cases}$$

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$$\begin{cases} f(x) - 1 = 2x f(x) + x g(x) \\ g(x) = x f(x) + x g(x) \end{cases}$$

This is a system of 2 eqns with 2 unknown functions $f(x)$ & $g(x)$. Upon solving it

$$f(x) = \frac{1-x}{x^2-3x+1}$$

$$g(x) = \frac{x}{x^2-3x+1}$$

Again

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$g(x) = \sum_{n=0}^{\infty} \frac{g^{(n)}(0)}{n!} x^n$$

but it is difficult to see a general pattern for $\frac{f^{(n)}(0)}{n!}$ & $\frac{g^{(n)}(0)}{n!}$

So again we use a trick based on partial fractions applied to $f(x)$ & $g(x)$ which luckily are rational functions

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So once Δ is computed for $x^2 - 3x + 1$
we get two roots (as $\Delta > 0$)

$$\gamma = \frac{3 + \sqrt{5}}{2}$$

$$\delta = \frac{3 - \sqrt{5}}{2}$$

see footnote:

and thus: $x^2 - 3x + 1 = (x - \delta)(x - \gamma)$ $a=1$

$$ax^2 + bx + c = 0 \quad = (\delta - x)(\gamma - x)$$

Partial fraction yields

$$f(x) = \frac{1-x}{x^2-3x+1} = \frac{A}{\gamma-x} + \frac{B}{\delta-x}$$
$$= \left(\frac{5+\sqrt{5}}{10} \right) \frac{1}{\gamma-x} + \left(\frac{5-\sqrt{5}}{10} \right) \frac{1}{\delta-x}$$

$$g(x) = \frac{x}{x^2-3x+1} = \frac{\tilde{A}}{\gamma-x} + \frac{\tilde{B}}{\delta-x}$$
$$= \left(\frac{-5-3\sqrt{5}}{10} \right) \frac{1}{\gamma-x} + \left(\frac{-5+3\sqrt{5}}{10} \right) \frac{1}{\delta-x}$$

Again upon using geometric series formula

$$\frac{1}{1-q} = 1 + q + q^2 + \dots + q^n + \dots \quad |q| < 1$$

Footnote: Note that as we try to solve a real
physical problems the roots may
be in an inconvenient form.

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We arrive at:

$$\begin{aligned}
 f(x) &= \sum_{n=0}^{\infty} \left[\frac{5+\sqrt{5}}{10} \frac{1}{8} \left(\frac{x}{8}\right)^n + \frac{5-\sqrt{5}}{10} \frac{1}{8} \left(\frac{x}{8}\right)^n \right] \\
 &= \sum_{n=0}^{\infty} \underbrace{\left[\frac{5-\sqrt{5}}{10} \left(\frac{3+\sqrt{5}}{2}\right)^{-(n+1)} + \frac{5-\sqrt{5}}{10} \left(\frac{3-\sqrt{5}}{2}\right)^{-(n+1)} \right]}_{\hat{a}_n} x^n \\
 &= \sum_{n=0}^{\infty} \hat{a}_n x^n
 \end{aligned}$$

$$\begin{aligned}
 g(x) &= \sum_{n=0}^{\infty} \left[\frac{-5-3\sqrt{5}}{10} \frac{1}{8} \left(\frac{x}{8}\right)^n + \frac{-5+3\sqrt{5}}{10} \frac{1}{8} \left(\frac{x}{8}\right)^n \right] \\
 &= \sum_{n=0}^{\infty} \left[\frac{-5-3\sqrt{5}}{10} \left(\frac{3+\sqrt{5}}{2}\right)^{-(n+1)} + \frac{-5+3\sqrt{5}}{10} \left(\frac{3-\sqrt{5}}{2}\right)^{-(n+1)} \right] x^n \\
 &= \sum_{n=0}^{\infty} \hat{b}_n x^n
 \end{aligned}$$

Thus $\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \hat{a}_n x^n$

$$\Downarrow \\
 \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} \hat{b}_n x^n$$

$$\Downarrow \\
 b_n = \hat{b}_n$$

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Hence

$$\begin{cases} a_n = \frac{5+\sqrt{5}}{10} \left(\frac{3+\sqrt{5}}{2} \right)^{-(n+1)} + \frac{5-\sqrt{5}}{10} \left(\frac{3-\sqrt{5}}{2} \right)^{-(n+1)} \\ b_n = \frac{-5-3\sqrt{5}}{10} \left(\frac{3+\sqrt{5}}{2} \right)^{-(n+1)} + \frac{-5+3\sqrt{5}}{10} \left(\frac{3-\sqrt{5}}{2} \right)^{-(n+1)} \end{cases}$$

Note that if $f(x) = \frac{W_n(x)}{W_m(x)}$ $\deg(V_n(x)) = n$
 $\deg(W_m(x)) = m$

then each polynomial can be factorized as a multiplication of quadratic & linear polynomials (this requires root finding for $W_m(x)$)

One can apply method of generating function to schemes of non-fixed orders e.g.

$$t(n) = t(0)t(n-1) + t(1)t(n-2) + \dots + t(n-1)t(1)$$

and exploit Cauchy formula for series multiplication. To be discussed later.